Synthetic OD Matrix Generation: Problem Statement and Constraints

OD Estimation Project

July 22, 2026

Abstract

This note states the problem of generating realistic Origin–Destination (OD) matrices from first principles for machine-learning training. It specifies inputs and outputs, hard and soft transportation constraints, desired statistical properties, and the requirement that samples lie on a low-dimensional latent manifold rather than filling the full N^2-dimensional space uniformly.

1 Problem Statement

1.1 Objective

Develop a statistical generator capable of producing an arbitrary number of realistic Origin–Destination (OD) matrices for a transportation network.

Each generated OD matrix should satisfy fundamental transportation principles and network constraints while exhibiting sufficient variability to serve as high-quality training data for machine learning models.

Unlike traditional approaches that perturb an existing OD matrix, the proposed generator should synthesize OD demand **from first principles**, producing diverse yet realistic travel demand patterns.

1.2 Inputs

The generator requires only:

- Transportation Analysis Zones (TAZs)
- Pairwise distance (or travel cost/time) matrix
- Desired total number of trips
- Number of OD matrices to generate

No historical OD matrix is required.

1.3 Output

Generate

$$\{T^{(1)}, T^{(2)}, \ldots, T^{(K)}\}$$

where

$$T^{(k)} \in \mathbb{R}^{N\times N}$$

is a valid OD matrix.

2 Constraints

2.1 Non-negativity

Every OD flow represents trips. Therefore,

$$T_{ij} \geq 0.$$

2.2 No Intrazonal Trips (Optional)

If intrazonal trips are excluded,

$$T_{ii} = 0.$$

Otherwise, they may be permitted.

2.3 Fixed Total Demand

Every generated matrix should represent the same total daily travel demand:

$$\sum_{i,j} T_{ij} = G,$$

where G is a user-specified constant.

2.4 Production–Attraction Consistency

The total number of productions must equal the total number of attractions:

$$\sum_i P_i = \sum_j A_j = G,$$

where

$$P_i = \sum_j T_{ij}, \qquad A_j = \sum_i T_{ij}.$$

If production and attraction vectors are prescribed, every generated matrix must satisfy them exactly. Otherwise, these vectors are generated as part of the synthesis process.

2.5 Distance Decay

Trips generally decrease with increasing spatial separation between origin and destination. For two OD pairs with

$$d_{ij} > d_{mn},$$

the expected flow should satisfy

$$\mathbb{E}[T_{ij}] < \mathbb{E}[T_{mn}].$$

However, this is **not** a hard constraint. Long-distance trips must still occur with non-zero probability. Thus,

$$T_{ij} \propto f(d_{ij}),$$

where $f(d)$ is a monotonically decreasing function such as

- exponential decay,
- power-law decay,
- logistic decay,
- gravity-based decay.

The distance decay should be interpreted statistically rather than deterministically. While nearby zones generally exchange more trips, some long-distance OD pairs should still exhibit high demand due to latent socioeconomic or transportation factors.

2.6 Random Corridor Dominance

Each generated OD matrix should exhibit a unique set of dominant travel corridors. Specifically:

- A small fraction of OD pairs should carry very high demand.
- Most OD pairs should carry moderate demand.
- Many OD pairs should carry low demand.

The identity of these dominant corridors should vary from one generated matrix to another. The generator must therefore avoid repeatedly producing the same dominant OD pairs.

2.7 Heavy-Tailed Flow Distribution

Cell values should approximately follow a heavy-tailed distribution. Consequently:

- Most OD pairs should carry small demand.
- A few OD pairs should carry very large demand.

This reproduces empirical transportation demand patterns observed in real road networks.

2.8 Approximate Reciprocity (Symmetry)

Daily travel demand often exhibits reciprocal movement. Therefore,

$$T_{ij} \approx T_{ji}.$$

Exact equality is unnecessary. The generator should preserve only statistical symmetry while allowing directional imbalances.

2.9 Spatial Coherence

Nearby OD pairs should exhibit correlated behavior. If one OD pair forms a major travel corridor, neighboring origins or destinations should have an increased probability of generating similar travel demand. Demand should therefore appear spatially coherent rather than as isolated random spikes.

2.10 Smooth Origin and Destination Importance

Each origin possesses a production strength. Each destination possesses an attraction strength. Zones with larger production strengths tend to generate more trips; zones with larger attraction strengths tend to receive more trips. These strengths should vary across generated matrices.

2.11 Diversity

Generated matrices should not be nearly identical. For two generated matrices $T^{(a)}$ and $T^{(b)}$, require

$$\|T^{(a)} - T^{(b)}\|_F > \tau,$$

where τ is a predefined diversity threshold.

2.12 Entropy

The generator should maximize dataset diversity while maintaining transportation realism. Different OD cells should experience varying demand distributions across the generated dataset.

2.13 Sparsity

Depending on the intended application, many OD pairs should contain zero, negligible, or low flow. Completely dense OD matrices are generally unrealistic.

2.14 Network Connectivity

Trips may only be assigned between connected origin–destination pairs. The generator should never allocate demand to disconnected zones.

2.15 Statistical Stationarity

All generated matrices should originate from the same underlying stochastic process. No generated matrix should be an obvious statistical outlier unless explicitly requested.

2.16 Scalability

The generator should efficiently produce 10^6 or more synthetic OD matrices without significant computational overhead.

3 Desired Statistical Properties

Across the generated dataset, the following properties should hold:

- Constant total demand.
- Realistic production distributions.
- Realistic attraction distributions.
- Heavy-tailed OD flow distribution.
- Distance-decay behavior.
- Approximate reciprocity.
- High entropy.
- High pairwise diversity.
- Spatial coherence.
- Stable marginal statistics.
- No duplicate matrices.
- Realistic trip-length distribution.
- Controllable randomness through a small number of interpretable parameters.

4 Additional Constraint: Low-Dimensional Latent Structure

Real-world OD matrices are not arbitrary. Instead, they are governed by a relatively small number of latent factors such as

- residential population,
- employment,
- commercial centers,
- educational institutions,
- industrial zones,
- recreational areas,
- accessibility,
- socioeconomic characteristics.

Consequently, although an $N \times N$ OD matrix contains N^2 entries, its effective dimensionality is considerably smaller.

A realistic generator should therefore produce OD matrices that lie on a low-dimensional manifold rather than uniformly occupying the entire N^2-dimensional space. This property explains why dimensionality reduction techniques such as Principal Component Analysis (PCA) and Autoencoders can effectively compress OD matrices while preserving most of their information.

5 Overall Problem Definition

Design a constrained probabilistic generator capable of sampling realistic Origin–Destination matrices from a transportation-aware probability distribution defined by

- network topology,
- distance-decay effects,
- production–attraction consistency,
- spatial coherence,
- approximate reciprocity,
- heavy-tailed corridor demand,
- dataset diversity,
- scalability.

The generated matrices should satisfy transportation constraints while remaining statistically diverse, physically plausible, and suitable for large-scale machine learning applications.